

UNIVERSITY OF THE PUNJAB

NOTIFICATION

It is hereby notified that the Vice Chancellor has approved the following recommendations/decisions made in a meeting of Centre for Quran and Sunnah chaired by the Pro-Vice Chancellor:-

کریڈٹ آورز (Credit Hours) کی تقسیم:

یونیورسٹی نوٹیفیکیشن کی روشنی میں بی ایس (آنرز) کے لیے سالانہ ایک کریڈٹ آور (Credit Hour) کرلیب ورک کریڈٹ آور کے مساوی قرار دیا گیا ہے۔ اور اس طرح ہر سمسٹر میں ہفتہ وار تین کلاسیں پڑھائی جائیں گی۔

پہلا، تیسرا، پانچواں اور ساتواں سمسٹر نان کریڈٹ آور (Non Credit Hour) تصور کیا جائے گا۔ جبکہ دوسرا، چوتھا، چھٹا اور آٹھواں سمسٹر (ہر سمسٹر) ایک کریڈٹ آور کے مساوی ہوگا۔

اس سہیت کی assessment باقی مضامین کی طرح ہوگی اور ان کے بارے میں مزید تفصیلی مسودہ باقی امتحان بھی باقی مضامین کی طرح ہوگا۔

نصاب:

الف۔ بی ایس آنرز :

سال اول : سمسٹر I - سورة الفاتحة سورة ال عمران -

سمسٹر II - سورة النساء سورة الأنعام -

سال دوم : سمسٹر III - سورة الأعراف سورة يونس -

سمسٹر IV - سورة هود سورة الكهف -

سال سوم : سمسٹر V - سورة مريم سورة النرقان -

سمسٹر VI - سورة الشعراء سورة ص -

سال چہارم : سمسٹر VII - سورة الزمر سورة ق -

سمسٹر VIII - سورة الذاریات سورة الناس -

ب۔ ایم اے/ایم ایس سی : بی ایس آنرز کے سال سوم اور چہارم کا نصاب

ج۔ ایم فل : سورة المؤمنون ، سورة الأحزاب اور سورة الحشر کا ترجمہ

دہی ایچ ڈی : سورة النور ، سورة الفرقان اور سورة الحجرات

نوٹ: i. ایم اے، ایم فل اور پی ایچ ڈی کا کورس ایک کریڈٹ آور (Credit Hour) پر مشتمل ہوگا۔

ii. بی ایس آنرز کے پہلے دو سال کا نصاب ADP میں پڑھایا جائے گا۔

تدریس ترجمہ قرآن کورس، بی ایس آنرز کے پہلے سمسٹر سے اٹھویں سمسٹر کے لیے
مندرجہ ذیل کورس کوثر ترتیب دیے ہیں۔

سال اول:

HQ-001	تدریس ترجمہ قرآن	سمسٹر I-
HQ-002	تدریس ترجمہ قرآن	سمسٹر II-

سال دوم:

HQ-003	تدریس ترجمہ قرآن	سمسٹر III-
HQ-004	تدریس ترجمہ قرآن	سمسٹر IV-

سال سوم:

HQ-005	تدریس ترجمہ قرآن	سمسٹر V-
HQ-006	تدریس ترجمہ قرآن	سمسٹر VI-

سال چہارم:

HQ-007	تدریس ترجمہ قرآن	سمسٹر VII-
HQ-008	تدریس ترجمہ قرآن	سمسٹر VIII-

Admin. Block,
Quaid-i-Azam Campus,
Lahore.

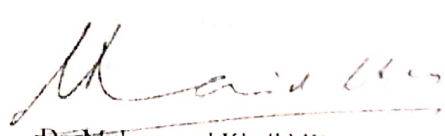
Sd/-
Dr. Muhammad Khalid Khan
Registrar

No. D/330/Acad.

Dated: 05-01-2021

Copy of the above is forwarded to the following for information and further
necessary action:-

1. Pro-Vice Chancellor
2. Deans of Faculties
3. Heads of University Teaching Departments
4. Secretary to the Vice-Chancellor
5. Private Secretary to Registrar
6. Notification file


Dr. Muhammad Khalid Khan
Registrar

UNIVERSITY OF THE PUNJAB
NOTIFICATION

It is hereby notified that the Syndicate in its meeting held on 18.08.2020 approved the recommendations of the Academic Council dated 24.06.2020 regarding teaching of Holy Quran with translation under a separate centre "Centre for Quran and Sunnah", as per following guidelines proposed by the Seven-Members Committee of Vice Chancellors constituted by the honourable Chancellor/Governor Punjab:-

1. The teaching of translation of the Holy Quran may be relevant to the profession, faith, ethics & morality and practices (ایمان، عبادت، معاملات، اخلاقیات و عادات).
2. The teaching of the Holy Quran may be undertaken with "translation only".
3. The Quran course may have equivalent to one credit hour per year (Theory/Practical), as the relevant University feels appropriate. However, the HCE limit of total credit hours in Undergraduate Program must be followed.
4. The Quran course may be taught in addition to the HEC-approved compulsory Pakistan and Islamic studies.
5. The assessment of the Quran course must require each candidate to pass/fail. The student may be further assigned a grade as in any other credited subject, if University feels necessary/appropriate.
6. The Quran course is mandatory/compulsory and may be spread in a manner to engage students throughout the duration of the course.
7. As a substitute, the non-Muslims may be offered the course of Ethics as is already the case in many universities.
8. Universities may make a concrete effort to generate interest in this compulsory course so that the course is taught in the best spirit of drawing learnings from the Holy Quran.
9. A suggested list of Quran Translations (Urdu & English) is attached.

Quaid-i-Azam Campus,
Lahore.

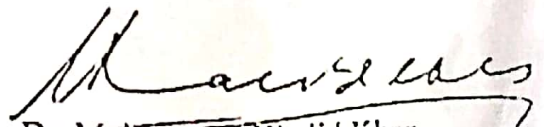
Sd/-
Dr. Muhammad Khalid Khan
Registrar

No. D/5477/Acad.

Dated: 01-09-2020.

Copy of the above is forwarded to the following for information and further necessary action:-

1. Pro-Vice Chancellor
2. All the Deans of the Faculties
3. All the Heads of the University Teaching Departments
4. Controller of Examinations
5. Director, IT Centre
6. Deputy Registrar (Affiliation)
7. Secretary to the Vice-Chancellor
8. Secretary to the Registrar
9. Admin. Officer (Statutes)
10. Admin. Officer (Senate/Syndicate)
11. Admin. Officer (Meetings-II)


Dr. Muhammad Khalid Khan

Model Paper for Affiliated Colleges



Course:
Program:
Duration:
Paper Date:

Translation of Holy Quran
BS 4 Year Degree Program
Two Hours

Department:
Exam:

Final Term Exam.

Course Code: HQ-001
Semester: Spring/Fall 20--
Total Marks: 60
Page(s): 3
Student's Name: _____
Roll No: _____

Instruction/Notes:

Attempt this Paper on the Same Sheet

20 نمبر

سوال 1 : مندرجہ ذیل آیات مبارکہ کا اردو/انگریزی میں ترجمہ کریں۔

Translate the Following Verses into Urdu or English Language

1- حَافِظُوا عَلَى الصَّلَوَاتِ وَالصَّلَاةِ الْوُسْطَى وَقُومُوا لِلَّهِ قَانِتِينَ.

2- وَقَاتِلُوا فِي سَبِيلِ اللَّهِ وَعَلِمُوا أَنَّ اللَّهَ سَمِيعٌ عَلِيمٌ.

3- قَالَ إِنَّ اللَّهَ اصْطَفَاهُ عَلَيْكُمْ وَزَادَهُ بَسْطَةً فِي الْعِلْمِ وَالْجِسْمِ وَاللَّهُ يُؤْتِي مَلَكُهُ مَن يَشَاءُ وَاللَّهُ وَاسِعٌ عَلِيمٌ

4- قَوْلٌ مَعْرُوفٌ وَمَغْفِرَةٌ خَيْرٌ مِّنْ صَدَقَةٍ يَتْبَعُهَا أَدَى ۖ وَاللَّهُ غَنِيٌّ حَلِيمٌ

5- يَا أَيُّهَا الَّذِينَ آمَنُوا لَا تُبْطِلُوا صَدَقَاتِكُمْ بِالْمَنِّ وَالْأَذَى كَالَّذِي يُنْفِقُ مَالَهُ رِثَاءَ النَّاسِ وَلَا يُؤْمِنُ بِاللَّهِ وَالْيَوْمِ الْآخِرِ ۖ

6- نَزَّلَ عَلَيْكَ الْكِتَابَ بِالْحَقِّ مُصَدِّقًا لِّمَا بَيْنَ يَدَيْهِ وَأَنزَلَ التَّوْرَةَ وَالْإِنْجِيلَ

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7- رَبَّنَا لَا تُزِغْ قُلُوبَنَا بَعْدَ إِذْ هَدَيْتَنَا وَهَبْ لَنَا مِنْ لَدُنْكَ رَحْمَةً إِنَّكَ أَنْتَ الْوَهَّابُ

8- تَوَلَّجَ اللَّيْلُ فِي النَّهَارِ وَتَوَلَّجَ النَّهَارُ فِي اللَّيْلِ وَتُخْرِجُ الْحَيَّ مِنَ الْمَيِّتِ وَتُخْرِجُ الْمَيِّتَ مِنَ الْحَيِّ وَتَرْزُقُ مَنْ تَشَاءُ بِغَيْرِ حِسَابٍ

9- وَادْكُرْ رَبَّكَ كَثِيرًا وَسَبِّحْ بِالْعَشِيِّ وَالْإِبْكَارِ

10- وَمَنْ يَبْتَغِ غَيْرَ الْإِسْلَامِ دِينًا فَلَنْ يُقْبَلَ مِنْهُ وَهُوَ فِي الْآخِرَةِ مِنَ الْخَاسِرِينَ

10 نمبر

سؤال 2: ”رب“ کے ساتھ ضمائر متصلہ لگائیں اور ان کے معانی بتائیں۔

Use attached pronouns with ”رب“ and give their meanings.

- 1- رَبُّ + هُ =
.....
- 2- رَبُّ + هُمْ =
.....
- 3- رَبُّ + هَا =
.....
- 4- رَبُّ + هُنَّ =
.....
- 5- رَبُّ + كُ =
.....
- 6- رَبُّ + كُمْ =
.....

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- = رَبُّ + كِ = -7
- = رَبُّ + كُنْ = -8
- = رَبُّ + ي = -9
- = رَبُّ + نَا = -10

سوال 3 : مندرجہ ذیل قرآنی الفاظ کے اردو/انگریزی میں معانی لکھیں۔ 10 نمبر

Write the meanings of the following Qur'anic words

- 1 فِتْنَةٌ
-2 سَبِيلٌ
-3 ذَهَبٌ
-4 فِضَّةٌ
-5 نِسَاءٌ
-6 اَنْحَارٌ
-7 الْمَلَكُوتِ
-8 قَسِطٌ
-9 سَمِيعٌ
-10 ذُرِّيَّةٌ
-11 عَاوِرٌ
-12 مَهْدٌ
-13 يَنْبُرُكُ
-14 طَيْرٌ
-15 يَنْتٌ
-16 تَاْكُلُوْنَ
-17 نَحْلٌ
-18 الَّذِيْنَ
-19 طِفَامٌ
-20 بَرٌ

10 نمبر

سوال 4 : مندرجہ ذیل الفاظ کی مدد سے واحد کی جمع لکھیں۔

Write the plural with the given words.

أُولَئِكَ - الَّذِينَ - مَوَاتٍ - هُمْ - يَنْتَظِرُونَ - كُتُبٌ - غَافِلُونَ - آيَاتٌ -

يُؤْتِ - أَبْوَابٌ - مُحْسِنِينَ - أَلْبَابٌ - أَعْمَالٌ - يَتْلُو - مُؤْمِنُونَ - تَوَابِينَ - أَصْحَابٌ -

ذَلِكَ	هُوَ	يَنْتَظِرُونَ	كِتَابٌ	بَابٌ	
الَّذِينَ	يَنْتَظِرُونَ	إِلَهُ	غَافِلُونَ	مُحْسِنِينَ	
مَوَاتٍ	مُؤْمِنُونَ	يَدٌ	صَاحِبٌ	لُبٌ	
يَتْلُو	أَعْمَالٌ	رَسُولٌ	آيَةٌ	تَوَابٌ	

10 نمبر

سوال 5 : آیات کو ان کے ترجمے سے لکیر کے ذریعے ملائیں۔

Match the verses with lines to the translation.

ہم صرف تیری ہی عبادت کرتے ہیں اور ہم
صرف تجھ سے ہی مدد مانگتے ہیں۔

1 وَلَا تَلْبِسُوا الْحَقَّ بِالْبَاطِلِ وَتَكْتُمُوا
الْحَقَّ وَأَنْتُمْ تَعْلَمُونَ۔

اس نے کہا میں اللہ کی پناہ میں آتا ہوں کہ میں
جاہلوں میں سے ہو جاؤں۔

2 إِيَّاكَ نَعْبُدُ وَإِيَّاكَ نَسْتَعِينُ۔

(یہ) وہ کتاب ہے جس میں کوئی شک نہیں،
پرہیزگاروں کے لیے ہدایت ہے۔

3 قَالَ أَعُوذُ بِاللَّهِ أَنْ أَكُونَ مِنَ الْجَاهِلِينَ

اور حق کو باطل کے ساتھ خلط ملط نہ کرو، اور حق
کو (ن) چھپاؤ حالانکہ تم جانتے ہو۔

4 قُلْ لِلَّهِ الْمَشْرِقُ وَالْمَغْرِبُ

آپ کہہ دیجئے (کہ) مشرق اور مغرب اللہ ہی کے
لیے ہے۔

5 ذَلِكَ الْكِتَابُ لَا رَيْبَ فِيهِ هُدًى
لِّلْمُتَّقِينَ۔



Code	Subject Title	Cr. Hrs	Semester
PST-111	Pakistan Studies	2	II
Year	Discipline		
1	Botany, Zoology, Mathematics-I,II, Statistics-I,II,III, Chemistry-I,II, Applied Psychology, Business Administration, Commerce, Economics, Sociology & Sociocultural Studies, Social Work, Political Science, Physics, Mass Communication, Information Technology, Islamic Education, History, Education (Elementary), Education (Secondary), Urdu		

1. Pakistan Movement: Historical and Ideological Perspective

- Muslim Revivalist Movements in India
- The Two-Nation Theory: From Sir Syed Ahmed Khan to Allama Iqbal
- Quaid-i-Azam Muhammad Ali Jinnah and the Struggle for Pakistan Creation of Pakistan: Early Challenges

2. Land and People of Pakistan

- Physical features and Geo-Strategic Location of Pakistan
- Culture and Traditions: Regional Dimensions
- Social Issues and Challenges to Pakistan

3. History and Politics in Pakistan (1947-2008)

- Early Parliamentary Phase
- Ayub and Yahya's Era
- Democratic Era (1971-1977)
- Zia Era (1977-88)
- Civil Rule (1988-99)
- Musharaf Era (1999-2008)

4. Contemporary Pakistan

- The Constitution of 1973: Salient Features
- Foreign Policy and Relations with neighboring countries
- Salient Features of Economy, Agriculture and Industry, National Resources

Recommended Books:

1. Muhammad Raza Kazni, Pakistan Studies Core Texts for Colleges and universities, Oxford University Press (2006)
2. Ishtiaq Hussain Qureshi, The Struggle for Pakistan, Karachi, University of the Karachi, 1974
3. Khalid Bin Sayeed, Pakistan the Formative Phase (1857-1948), Karachi, Oxford University Press, 1968.
4. Choudhary M. Au, The Emergence of Pakistan, Lahore, Research Society of Pakistan, 2001
5. S. Qalb-i-Abid, Muslim Struggle for Independence (1857-1947), Sang-c-Med Publications, Lahore, 1997
6. M.R. Kazimi, A Concise History of Pakistan, Oxford University Press, 2009
7. Mehmood, Safdar. Pakistan Political Roots & Development. Lahore, 1994.
8. Afzal, M. Rafique, Pakistan: History and Politics 1947-1971, Karachi, Oxford University Press, 2007



9. Rizvi, Hasan-Askari, The Military State and Society in Pakistan, Lahore, 2002
10. Burke, S.M., and Qureshi, S.A.D., The British Raj in India, Karachi, Oxford University Press, 1995
10. S.M. Burke and Lawrence Ziring. Pakistan's Foreign policy: An Historical Analysis. Karachi: Oxford University Press, 1993.
11. Akbar, S. Zaidi, Issues in Pakistan's Economy, Karachi: Oxford University Press, 2000
12. Jinnah of Pakistan, Karachi, Oxford University Press, 1989
13. Zuifi Bhutto of Pakistan, New York, Oxford University Press, 1989
14. Shahid, M. Amin, Pakistan's Foreign Policy: A Reappraisal, (Second Edition), Karachi, Oxford University Press, 2010
15. Abdul Sattar, Pakistan's Foreign Policy: A Concise History 1947-2009 (2nd Edition), Karachi, Oxford University Press, 2010
16. Hardy, Peter, The Muslims of British India, New Delhi, Cambridge University Press, 1998





Code	Subject Title	Cr. Hrs	Semester
ENG-112	English-II (Academic Reading and Writing)	3	II
Year	Discipline		
1	Botany, Zoology, Mathematics-I,II, Statistics-I,II,III, Chemistry-I,II, Applied Psychology, Business Administration, Commerce, Economics, English, Sociology & Sociocultural Studies, Social Work, Political Science, Physics, Mass Communication, Islamic Education, History, Education (Elementary), Education (Secondary), Urdu		

Basics of Grammar II

Recommended Reading:

Eastwood, John. *Oxford Practice Grammar*. UK: Oxford University Press, 1992.

- Unit No. 4, 5, 8, 9, 11, 12, 13, 16, 18, 20, 22, 23, 24, 26, 28, 29, 30 (Tenses)
- Unit No. 128, 129, 130, 131 (Phrasal Verbs).

Reading and Critical Thinking

Recommended Reading:

Ellison, Peter and Gould, Mike. *Focus on Comprehension*. Volume 4. Singapore: Learner's Publishing Pte, 2009.

- Section 4: Unit No. 12, 13, 14, 15. Practice Unit 3.
- Section 5: Unit No 16, 17, 18.
- Section 6: Unit No. 19. Practice Unit 4.

Paragraph Writing II

The students are required to know basics of Paragraph Writing with an emphasis on Topic sentences and Supporting sentences and a possible Concluding sentences. (Word Limit Up to 120 words)

Recommended Reading:

Arnaudet, Martin L. and Barrett, MaryEllen. *Paragraph Development: A Guide for Students of English as a Second Language*. Prentice-Hall, Englewood Cliff, N.J. 1981. (Page 179-185).

Study Skills

The students are expected to be proficient in Reading Skills like Skimming, Scanning, Speed Reading and avoiding Faulty Reading Habits.

Recommended Reading:

Aziz, Nadeem. *English Language Communication Skills for B.Ed*. Pakistan: Majeed Book Publishers, (Page 139-159).

Vocabulary Building Skills

WORD ROOT METHOD Unit 1-6. Page No. 88-103.

Recommended Reading:

Idrees, Muhammad. *Guide for GAT General Test. SMART BRAIN. GRE (General, Local)* 2010-2011 Ed. Lahore: Dogar Brother Publishers, 2010.



Note: The concepts listed in the syllabus contents may be acquired from sources other than those recommended.





Code	Subject Title	Cr. Hrs	Semester
MATH-103	Mathematics A-II [Plane Curves & Analytic Geometry]	4	II
Year	Discipline		
1	Mathematics-I,II, Chemistry-II, Statistics-I,II,III		

Plane Analytics Geometry

- Conic section and quadratic equations
- Classifying conic section by eccentricity
- Translation and rotation of axis
- Properties of circle, parabola, ellipse, hyperbola
- Polar coordinates, conic sections in polar coordinates
- Graphing in polar coordinates
- Tangents and normal, pedal equations, parametric representations of curves

Applications of Integration

- Asymptotes.
- Relative extrema, points of inflection and concavity
- Singular, points, tangents at the origin
- Graphing of Cartesian and polar curves
- Area under the curve, area between two curves
- Arc length and intrinsic equations
- Curvature, radius and centre of curvature
- Involute and evolute, envelope

Analytic Geometry of Three Dimensions

- Rectangular coordinates system in a space
- Cylindrical and spherical coordinate system
- Direction ratios and direction cosines of a line
- Equation of straight lines and planes in three dimensions
- Shortest distance between skew lines
- Equation of sphere, cylinder, cone, ellipsoids, paraboloids, hyperboloids
- Quadric and ruled surfaces
- Spherical trigonometry. Direction of Qibla

Recommended Books

1. Thomas, *Calculus*, 11th Edition. Addison Wesley publishing company, 2005
2. H. Anton, I. Bevens, S. Davis, *Calculus*, 8th Edition, John Wiley & Sons, Inc. 2005
3. Hughes-Hallett, Gleason, McCallum, et al, *Calculus Single and Multivariable*, 3rd Edition. John Wiley & Sons, Inc. 2002.
4. Frank A. Jr, Elliott Mendelson, *Calculus*, Schaum's outlines series, 4th edition, 1999
5. C.H. Edward and E.D Penney, *Calculus and Analytics Geometry* Prentice Hall, Inc. 1988
6. E. W. Swokowski, *Calculus and Analytic Geometry* PWS Publishers, Boston, Massachusetts, 1983.
7. Dennis G. Zill & Patric D. Shanahan, *Complex Analysis*, Jones & Barlett Publishers, 2003



Code	Subject Title	Cr. Hrs	Semester
COMP-103	Computer-II	3	II
Year	Discipline		
1	Statistics-II, Islamic Education		

- 1. **Introduction:**
 - a. Data, Processing & Information
 - b. Data Item/Entity, Record, File, Metadata, Database
 - c. Files and its types
 - d. Traditional File Processing Systems and its disadvantages
- 2. **Database**
 - a. Database Systems
 - b. Database Objectives
 - c. Database Management Systems
 - d. Components of the DBMS
 - e. Types of DBMS
 - f. Database Applications
 - g. Advantages of Database System
 - h. Disadvantages of Database System
- 3. **Database Development**
 - a. Database development with information system
 - b. Database development process
 - c. 3-Schema Architecture
- 4. **Data Associations**
 - a. Entity, Attribute & Associations
 - b. Entity, Attribute & Associations
 - c. Types of Associations
 - d. Entity-Relation Model
- 5. **Relation & its Characteristics**
 - a. Relation
 - b. Properties of Relations
 - c. Key and its types
- 6. **Normalization**
 - a. Normalization and its steps
 - b. Types of Normalization
 - c. Effects for Normalization
- 7. **Database Design**
 - a. Logical Database Design
 - b. Physical Database
 - c. Database Administration



Code	Subject Title	Cr. Hrs	Semester
COMP-104	Computer Lab-II	1	II
Year	Discipline		
1	Statistics-II		

1. Database Implementation

- Using access
- Creating relations
- Forming attributes and fields
- Selecting record by using query
- Command introduction
- DBMS introduction in market

2. SQL

- Structured query language of relational DBMS
- Data definition language (DDL)
- Data manipulation Language (DML)
- DCL

3. DBMS Usage:

- How to switch among database management systems
- Similarities and dissimilarities of DBMS
- Interacting querying, updating, manipulating DBMS
- Maintenance of Database Management System

4. Queries

- Introduction of query
- Writing query
- Sub query
- Joins and queries

5. Practical Issues:

- Database connectivity and its issues (ODBC)
- Two tier and thee tier architecture issues
- Business rules and constraints and implementation in DBMS
- DBA

Recommended Books:

- Discovering Computers by Shelly Cashman, Computer application in business by Dr. Liaqat
- Peter Norton's Introduction to Computers, 6/e
- Microsoft Office 2003: The Complete Reference by Jennifer Kettell, Guy Hart-Davis and Curt Simmons
- Computer Fundamentals (Paperback) by P. K. Sinha
- Discovering Computers 2008: Living in a Digital World, Fundamentals (Shelly Cashman Series) by Gary B. Shelly and Misty E. Vermaat



Code	Subject Title	Cr. Hrs	Semester
STAT-103	Statistics-II	3	II
Year	Discipline		
1	Statistics-I,II,III, Mathematics-II, Economics		

Code	Subject Title	Cr. Hrs	Semester
STAT-104	Statistics Lab-II	1	II
Year	Discipline		
1	Statistics-I,II,III, Mathematics-II, Economics		

Course Outline

Random experiments, sample space and events. Counting techniques. Definitions and axioms of probability. Basic laws of probability. Independence of events. Bayes Theorem (proof and required) and its application.

Discrete Random Variable and Discrete Probability Distributions

Random variable, distribution function, discrete random variable. Probability distribution of a discrete random variable. Joint distribution of two discrete random variables, marginal and conditional distributions, mathematical expectation and its properties, mean, variance and moments. Concept of m.g.f. and its properties. Uniform, Bernoulli, Binomial, Hypergeometric and Poisson distributions, mean, variance and shape of these distributions and their properties. Application of these distributions with examples from various fields. Multinomial distribution (only application).

Continuous Random Variables & Continuous Probability Distributions

Continuous random variables. Probability distribution of a single continuous random variable, probability density function and distribution function. Mean, variance and moments of continuous random variables. Uniform and Normal distribution. Mean, variance and shape of these distributions and their properties. Application of these distributions. Normal approximation to the Binomial and Poisson distribution (just application). Fitting of Normal distribution by area method.

Recommended Books

1. Chaudhry, S.M. & Kamal, S. (2010). Introduction to Statistical Theory Part I, Ilmi Kitab Khana, Urdu Bazar, Lahore.
2. Crawshaw, J and Chambers, J. (2001). A concise course in advanced level Statistics with worked examples, Nelson Thornes, 4th Edition.
3. Graybill, Iyer & Burdick (1998). Applied Statistics, A first course in inference. Prentice Hall, New Jersey.
4. Beg, M.A. and Mirza, M.D. (2006). Statistics, Theory and Methods, Volume I, Carven Book House, Kutechery Road, Lahore.
5. Chase W. & Bown F. (1997). General Statistics, 3rd Edition, John Willy & Sons, New York.
6. Macfie, B.P. and Nufrio, P.M. (2006). Applied Statistics for public policy, Prentice Hall of India.
7. Blumen (1997), Elementary Statistics, 3rd Edition, McGraw Hill, New York.



8. Johnson, R.A. and Wichern, D.W. (2003). Business Statistics: Decision making with data, John Wiley & Sons Inc.
9. Levine, D.M., Kschbiel, T.C. and Berenson, M.L. (2003). Business Statistics: A first course, 3rd edition, Pearson Education.
10. Levin, J. and Fox, J.A. (2006). Elementary Statistics in Social Research, 10th edition, Pearson Education.
11. Medhi, J. (1992). Statistical Methods: An Introductory text, New Age International Publishers.
12. Chaudhry, R.M. (1998). Polymer Modern Statistics, Polymers.

